Trioron 1.1: a multi-axis substrate for resource-bounded continual learning

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Abstract

We describe *Trioron 1.1*, a substrate-level revision to the trioron architecture introduced in the companion paper, exposing six per-cell axes of structural plasticity: long-range input reach, plastic fanout sparsity, depth growth, slow axonal gain, dendritic compartmentalization, and field-conditional layer emergence. The six axes share one design principle — each is a separate write channel into the substrate, with its own provenance and its own plasticity gate, rather than a feature of the loss or the optimizer. A small spatial primitive (positions on a unit square, partitioned into pools) then enables a cell-granularity composition operator: two organisms trained independently on disjoint task slices can be absorbed into one substrate without retraining when they share the L0 perception layer's deterministic factor (the R·S handshake). The cell-granularity composition is compatible with positional locality constraints (locally-connected masks) because absorption respects pools, not exact positions. We report the mechanism's first verification on a chained-15 disjoint-class slice: task-aware accuracy is preserved on both donors within $\Delta \leq 0.026$, full-softmax accuracy lifts by $+0.20$ to $+0.31$ via manifold merge plus a 200-step head-only calibration pass, and the post-absorption substrate is *more* locality-clean than either donor alone. The 1.1 substrate is backwards-compatible with 1.0 donors: legacy checkpoints load unchanged and run under the extended forward path with the new axes at their default-off settings.

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